

P100 Unit Diagnostic Report

Modem Failure - Sent for Repair

Node (planename)	N010AV
Airline	AVA (Avianca)
Serial Number	W14031868522
Modem Model	Sierra Wireless EM7565
Modem Firmware	SWI9X50C_01.14.02.00 (correct, not the root cause)
Diagnostic Date	August 20, 2026
Diagnosis	Hardware failure - SIM interface controller
Modem State	state: failed / failed reason: sim-missing

Diagnostic Summary

Unit N010AV has a Sierra Wireless EM7565 modem that fails to detect the SIM card, consistently reporting failed reason: sim-missing. The modem enumerates correctly over USB, firmware is at the correct version (SWI9X50C_01.14.02.00), and the operating system's serial ports/symlinks are healthy. Software causes were ruled out through the troubleshooting steps detailed below, leading to the conclusion that the failure is hardware-related (SIM interface controller) - the same pattern previously confirmed on unit N096AV.

Troubleshooting Steps Performed (in order)

1. Verification of modem status via ModemManager (mmcli -m 0): confirms modem detected over USB with correct firmware, but reports state 'failed' with reason 'sim-missing'.
2. Physical inspection: the SIM card was removed, inspected, and reseated firmly in its slot. The issue persisted with no change.
3. Verification of system symlinks (/dev/modemPPP, /dev/modemAT): both present and correctly pointing to their corresponding ttyUSB ports - rules out a udev rules or USB enumeration issue.
4. Full modem reset (mmcli -m 0 --reset): after the reset and reinitialization (during which ModemManager reassigned the modem's internal index from 0 to 1, a normal side effect of the reset), the modem reports the exact same state (failed / sim-missing) - rules out a stuck QMI session or a transient initialization issue.

Conclusion: with modem hardware, firmware, system symlinks, and the physical SIM ruled out as the cause through the steps above, and no recovery after a full modem reset, the failure is attributed to the SIM interface controller of the Sierra Wireless EM7565 modem. Modem replacement is recommended.

Appendix: Technical Evidence (logs and command output)

Unit Identification

N010AV
AVA
Serial Number: W14031868522
UUID: 03000200-0400-0500-0006-000700080009
Wake-up Type: Power Switch
SKU Number: SGP100-713020-00;M0;W0

Modem Status (mmcli -m 0) - before reset

General | path: /org/freedesktop/ModemManager1/Modem/0
| device id: f8a21da68e99b83faeea65ad432c4e31177d4dd7
Hardware | manufacturer: Sierra Wireless, Incorporated
| model: EM7565
| firmware revision: SWI9X50C_01.14.02.00 2e210b jenkins 2020/08/19 14:18:39
| h/w revision: 1.0
| equipment id: 353533101469602
System | device: /sys/devices/pci0000:00/0000:00:15.0/usb1/1-4/1-4.1
| drivers: qmi_wwan, qcserial
| ports: cdc-wdm0 (qmi), ttyUSB3 (gps), wwan0 (net)
Status | state: failed
| failed reason: sim-missing
| power state: on
| signal quality: 0% (cached)
SIM | sim slot paths: slot 1: none (active)
| slot 2: none

System Symlinks

lrwxrwxrwx 1 root root 7 Aug 20 15:20 /dev/modemAT -> ttyUSB3
lrwxrwxrwx 1 root root 7 Aug 20 15:20 /dev/modemPPP -> ttyUSB4

Modem Status (mmcli -m 1) - after mmcli --reset

Note: ModemManager reassigned the modem's index from 0 to 1 after the reset - a normal side effect, not an error.

General | path: /org/freedesktop/ModemManager1/Modem/1
| device id: f8a21da68e99b83faeea65ad432c4e31177d4dd7
Hardware | manufacturer: Sierra Wireless, Incorporated
| model: EM7565
| firmware revision: SWI9X50C_01.14.02.00 2e210b jenkins 2020/08/19 14:18:39
Status | state: failed
| failed reason: sim-missing
| power state: on
SIM | sim slot paths: slot 1: none (active)
| slot 2: none

(No change from the pre-reset state)